

GE23311 - ENVIRONMENTAL SCIENCE AND SUSTAINABILITY

UNIT I ENVIRONMENT AND BIODIVERSITY

Syllabus: *Definition, scope and importance of environment – need for public awareness. Ecosystem and Energy flow– ecological succession. Types of biodiversity: genetic, species and ecosystem diversity– values of biodiversity, India as a mega-diversity nation – hot-spots of biodiversity – threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – endangered and endemic species of India – conservation of biodiversity: In-situ and ex-situ.*

Environment: *The sum of total of all living and non-living things around us influencing one another is called **Environment**.*

Environmental Science: *The study of environment, its biotic and abiotic components and their interrelationship is called environmental science.*

Environmental Studies: *The process of educating people for preserving quality environment is defined as environmental studies.*

Environmental engineering is the application of engineering principles to the protection and enhancement of quality of the environment for the public health and welfare.

SCOPE OF ENVIRONMENTAL STUDIES:

Scopes:

1. To get an awareness and sensitivity to the total environment and its problems.
2. To know the necessity of the conservation of natural resources.
3. To motivate active participation in environmental protection and improvement.
4. Developing skills to identify and solve environmental related problems.
5. Ability to evaluate environmental programmes in terms of ecology, economy, etc

IMPORTANCE OF ENVIRONMENTAL STUDIES: All the industry is polluting one way or the other and none of them is safe. There is no zero-pollution industry. The air we breathe, the water we drink, the food we consume and the land we live on are all contaminated by industrial activities. Because of the lack of self-discipline and not worrying about our future generation, we are polluting our valuable resources. To solve the environmental related issues, the knowledge of environmental studies is very important.

Importance:

1. Through environmental studies, "People will be understanding the concept of need of development without destruction of the environment".
2. An environmental study has a direct relation to the quality of life we live.
3. Environmental studies relate to our surroundings which we observe daily but never think seriously.
4. Environmental studies develop a concern and respect for the environment.

TYPES OF ENVIRONMENTS

Environment is divided into two categories

- a) Natural Environment
 - It is characterized by natural components.
 - All biotic (living) and abiotic (non-living) components are created through a natural process.
 - **Example:** Soil, Water, Air, trees etc.
- b) Man-made Environment
 - Man is the most powerful environmental agent.
 - He modifies the environment using modern technologies, according to his needs to a great extent. Hence this environment is created by man.
 - Example: House, Roads, Parks etc.

Need for public awareness

- o Public awareness very essential to help understand pros and cons of environmental problems.
- o The United Nations Conference on Environment and Development held in Rio de Janeiro in 1992 and popularly known as Earth summit followed by the world summit on sustainable Development in 2002, have highlighted key issues of global environmental concern.
- o Environmental pollution cannot be removed by laws alone.
- o The proper implementation and especially public participation are important aspects.
- o Public participation is possible only when the public is aware about the

ecological and environmental issues.

- o A drive by the government to ban the littering of polythene cannot be successful until the public understands the environmental implications of the same.
- o The public has to be educated about the fact that if we are degrading our environment, we are actually harming ourselves.

Types of Public Participation:

- Watch dog **monitors government or institutions to ensure they follow laws properly**
- Pressure group **influences government policies to protect its interests.**
- Advisory Council **expert advice to the government or an organization.**
- Enforcing the environmental laws **Ensuring environmental rules are followed and punishing violators.**

CONCEPT OF AN ECOSYSTEM:

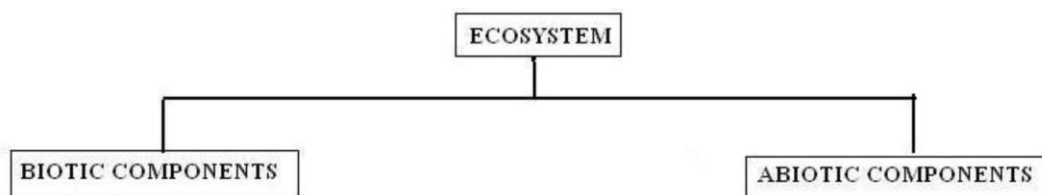
Ecology: The study of **interactions among the biotic and abiotic components in an ecosystem is called ecology.** **Ecosystem:** The **group of organisms interacting among themselves and with environment** is known as ecosystem. E.g. Forest ecosystem, marine ecosystem, etc.

STRUCTURE

The structure of an ecosystem explains the relationship between abiotic and biotic component. An ecosystem has two major components,

a) Abiotic (Non-living)

b) Biotic (Living)



- a) **Abiotic components:** Non-living components (physical and chemical) of an ecosystem collectively form a community called abiotic component.

Physical Component: It includes climate, energy, raw materials and living space that the **biological community needs.**

Chemical Component: It is the **sources of essential nutrients** such as protein, carbohydrate, N, O, K.

- b) **Biotic components:** Living members in an ecosystem collectively form a community is called biotic component. Based on the role in the ecosystem the biotic component classified into autotrophs and heterotrophs.

Autotrophic component: The members of autotrophic components are producers which are autotrophs. They derive energy from sunlight and make organic compounds from inorganic substances.

E.g. Green plants, algae etc.

Heterotrophic components: The members of heterotrophic components are consumers and decomposer.

Macro consumer: Herbivores, Carnivores and Omnivores. feed on plants or animals.

Micro consumer: Bacteria, Fungi etc. decompose dead plants and animals.

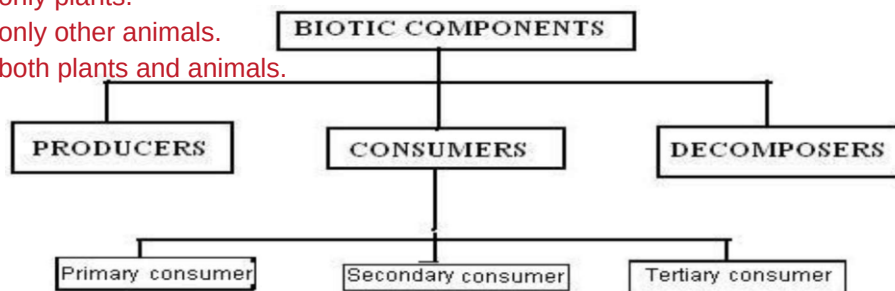
CLASSIFICATION OF BIOTIC COMPONENT:

The members of biotic component of an ecosystem are grouped into three groups based on how do they get food.

Herbivores: Animals that eat only plants.

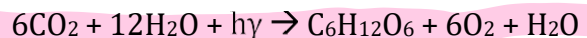
Carnivores: Animals that eat only other animals.

Omnivores: Animals that eat both plants and animals.



- a) **Producers:** Those organisms producing their own food through photosynthesis are called producers.

E.g. Plants, algae, etc



- b) **Consumers:** Those organisms which cannot prepare their food and depend directly or indirectly on the producer.

Primary consumer (Herbivores): They are directly depending on the plant for their food (Planteaters). E.g. Deer, Cow etc.

Secondary consumer (Primary carnivores): They directly depend on the primary consumers (Meateaters) E.g. Frog, cat, snake etc.

Tertiary consumer (Secondary carnivores): They directly depend on the secondary consumers (Meateaters) E.g. Lion, Tiger etc.

Omnivores: Those organisms which derive their energy from both producers and

consumers are called omnivores. E.g. Human being, crow, etc.

- c) **Decomposers:** Those organisms break down complex compounds of dead organisms are called decomposers. Ex. Bacteria and fungi.

FUNCTIONS OF AN ECOSYSTEM:

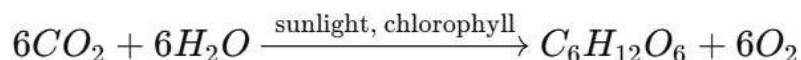
Primary Function: To synthesize the food (starch)

Secondary Function: To distribute the energy to all consumers or storage of energy by consumer.

Tertiary Function: The decomposition of dead organism.

ENERGY FLOW: Sun is the ultimate source of energy, its radiations travel through the space in the form of waves reaches the earth's atmosphere. It absorbs 50 % of the radiations and allows the remaining to reach the earth surface. Only 1 % of the sunlight is utilized for plant for photosynthesis.

The chemical equation for photosynthesis:



The producers are used by herbivores and herbivores are used by carnivores as their food. In this way energy is transferred from one organism to another and so on. This process is governed by law of thermodynamics,

I law of Thermodynamics:

"Energy can neither be created nor destroyed, but it can be converted from one form to another".

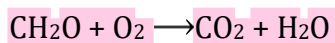
Solar energy $\rightarrow$ Chemical Energy in photosynthesis

II Law of Thermodynamics:

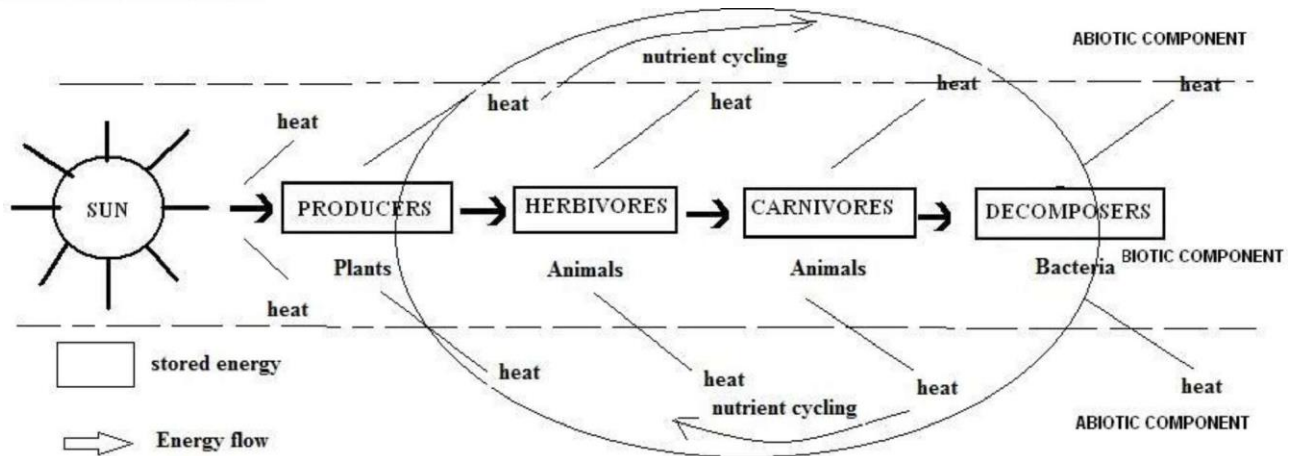
"Whenever energy is transformed, there is a loss of energy through heat".

This occurs when energy is transferred between trophic levels. The loss of energy takes place through respiration, running and hunting etc.

Respiration Equation:



From the above it is clear that, the biotic components and abiotic components are linked together through energy flow and nutrient cycling as shown in the following Fig.



ECOLOGICAL SUCCESSION:

The *progressive replacement of one community by another till the development of stable community in a particular area* is called ecological succession.

Stages:

Pioneer community – The first group of organism, which establish their community in the area.

Seres or Seral stage – Various developmental stages of the community in the area.

Types :-

Primary succession – It involves the gradual establishment of biotic community in a lifeless ground. The gradual development of a biotic community in a lifeless area where no life existed before.

Hydrarch – Establishment starts in a watery area.

Xerarch – Establishment starts in a dry area.

Secondary succession – Establishment of biotic communities in an area, where some type of biotic community is already present. The development of a biotic community in an area where life previously existed but was disturbed.

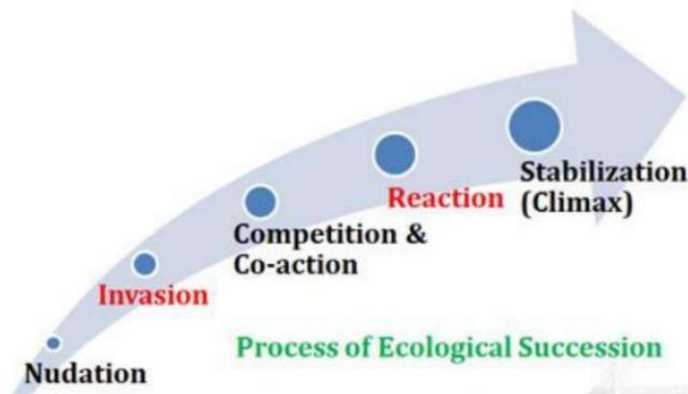
PROCESS OF ECOLOGICAL SUCCESSION:

The ecological succession is a complex process and it may take thousands of years. Frederic Clements in 1916 for the first time proposed the sequential phases of an ecological succession. The process of succession is completed through a series of sequential steps as given below:

- (1) Nudation
- (2) Invasion
- (3) Competition and Co-action

(4) Reaction

(5) Stabilization (climax)



1. **Nudation** – Nudation is the development of **bare area** in a lifeless area without any life form. It is the initial stage in ecological succession where a new area is exposed, lacking any living organisms. This could be due to natural events (like volcanic eruptions, floods, landslides, or erosion) or human-induced disturbances (like deforestation or agricultural practices).

Types of Nudation:

Nudation can be categorized as either primary or secondary. Primary nudation occurs on newly formed land (e.g., volcanic rock), while secondary nudation occurs on land where existing vegetation has been destroyed (e.g., after a forest fire).

2. Invasion:

Invasion is the successful establishment of a species in the bare area. It is the second step in ecological succession. A new species reaches the newly created bare area and they try to establish there (pioneer species).

The process of invasion is completed in THREE steps:

(A) Migration (Dispersal): Seeds, spores, propagules of a species reach the bare area due to migration. The migration can be achieved through air or water medium.

(B) Ecesis: It is the process of successful establishment of a species in the bare area.

The seeds or spores that reached the new area due to migration will germinate, grow and reproduce. Only a few progenies will survive due to the harsh environmental condition prevailing in the area.

(C) Aggregation: After ecesis, the individuals of a species increase their number and they stay close to each other. This process is called aggregation.

3. Competition

- Aggregation results in the increase of the number of species within a limited space.
- This results in competition between individuals for food and space.
- The competition may be intra-specific (individuals within a species) or inter-specific (individuals between species).
- Individuals of a species affect each other's life in various ways and this is called co-action.
- Competition and co-action results the survival of fit individuals and the elimination of unfit individuals from the ecosystem.
- A species with wide reproductive capacity and ecological amplitude only will survive.

4. Reaction – The living organisms take water, nutrients to grow and modify the environment is known as reaction.

- Reaction is the most important stage in the ecological succession. It is the modification of the environment through the influence of living organism present on it.
- Reaction cause change in soil, water, light and temperature of the area. Due to these modifications, the present community becomes unsuitable for the existing environmental conditions.
- Such communities will be quickly replaced by another community.
- The whole sequence of communities that replaces one another in the given area is called sere (sera).
- The various communities contributing sere are called seral communities or seral stages.

5. Stabilization – It leads to stable community, which is in equilibrium with the environment. It is the last stage in ecological succession.

- The final or terminal community becomes more or less stabilized for longer period of time. This community can maintain its equilibrium with the climate of the area.
- This final community is called the Climax Community (climax stage).

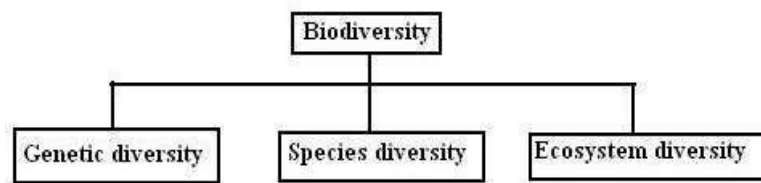
- The climax community is not immediately replaced by other communities.
- Climax community is determined by the climate of the region.
- Example of climax community: Forest, Grassland, Coral Reef.

BIO-DIVERSITY

Definition:

"The variety and variability among all group of organisms and the ecosystem in which they occur is called biodiversity".

Types of Biodiversity are as follows,



1. Genetic diversity

Genetic diversity is the diversity within the species. i.e variation of genes within the species. This is called genetic diversity. All rice species belong to the common species *oryza sativa*. But it has thousands of rice varieties which shows variation at the genetic level. E.g. **IR8, Basmathi, Ponni, Teak wood –Indian teak, burma teak** etc.

Example:

Garhwal Himalaya Moss Diversity: moss diversity changes with altitude and microhabitat.

A study in the Garhwal Himalaya investigated moss diversity across different altitudes and microhabitats. Researchers found that altitude and microhabitat significantly influenced moss species composition and distribution, with some species showing broad tolerance to different conditions.

2. Species diversity

Species diversity is the diversity between different species.

E.g. Plant species – Apple, Mango, Grape

Animal species - Lion, Tiger etc.

a. On farm crop diversity in Kerala:

A study focused on the on-farm conservation of plant genetic resources in Wayanad, Kerala. It highlighted the importance of traditional knowledge and cultural practices in preserving crop diversity, particularly among indigenous communities.

3. Ecosystem diversity

The diversity at the ecological or habitat level is called ecosystem diversity. E.g. River ecosystem, forest ecosystem.

Urban ecosystems:

A study on a university campus in India documented tree flora, revealing the diversity of tree species and their contribution to ecosystem services like food, medicine, and shade.

Marine ecosystems:

A case study on stopping oil exploration in the Tanon Strait in the Philippines highlights the importance of protecting marine biodiversity and the role of legal and community action in conservation.

VALUES OF BIODIVERSITY

As we get lot of benefits from other organism in number of ways. Sometime we realize the real value of the organism only after it is lost in the earth.

The value of Biodiversity can be mainly classified into

1. Direct Value

- i. Consumptive values
- ii. Productive use values

2. Indirect Value

- iii. Social values
- iv. Ethical values
- v. Aesthetic values
- vi. Optional values

i. Consumptive values

The consumptive value of biodiversity refers to **direct use of natural resources that are harvested and consumed without passing through a market.**

Example: Food, drugs and Fuel etc.

As food: Large number of wild plants consumed by human as food (Greens). Some of the wild animals are also consumed as food by Tribal people (Spider, Wild herbivores etc.)

As drugs: 70% of the modern medicines are derived from many plants. 20000 plants may be identified as a medicinal plant which are used in Siddha, Ayurvedha and Unani.

E.g. For example, the wonder drug penicillin used as an antibiotic is derived from a fungus called Penicillium, tetracycline from a bacterium.

Quinine from the bark of Cinchona tree (malaria), Morphine (Pain killer) and Reserphine (Hyper tension) are derived from valuable plants.

As Fuel: Firewood is directly used by the villages for cooking. Natural gas, petroleum and coal are also extracted from earth.

Consumptive use value is crucial for local communities, especially those who rely on natural resources for their livelihoods.

It highlights the direct economic benefits derived from biodiversity, even if these benefits are not always reflected in national economic statistics.

Understanding consumptive use value is important for sustainable resource management and conservation efforts.

ii. Productive use:

Productive use value of biodiversity refers to the **economic value derived from using natural resources**, such as timber, fish, and medicinal plants, that are harvested and sold in markets. The biodiversity products are obtained commercial values. The products are marketed and sold. They are derived from plants and animals.

E.g. Silk – silk worm, Wool – Sheep, Honey – Honey bee, etc Musk- musk deer, fur from many animals, lac from lac insects etc. are trades in market.

Besides, many industries are dependent upon productive use of values of biodiversity e.g.

paper and pulp industry, plywood industry, railway sleeper industry, textile industry, leather industry and pearl industry.

- iii. Social value:** It refers to the importance of biodiversity in people's social life, customs, religion, and spiritual beliefs.

Social value, in which social life, customs, religion and psycho-spiritual aspects of people are associated i.e. biodiversity has distinct social value, attached with different societies. It refers to the manner in which the bio resources are used to the society. Many plants and animals are considered as holy plants, holy animals in our country.

E.g. Holy plants - Tulsi, Neem, Lotus etc. Holy animals - Cow, Snake and Peacock etc.

- iv. Ethical value:** All life must be preserved. The biodiversity is considered to have a great value on religious and cultural basis. E.g. The river Ganges – Holy River, Neem, Tulsi are worshiped by Tamilians.

- v. Aesthetic value:** The beautiful nature of plant and animal insist us to protect biodiversity. This can be achieved by eco- tourism.

Eco-tourism - People from far palces, spend lot of money and time to vist the beautiful area and enjoyed the aesthetic value. E.g: Pleasant music of birds, colours of butterfly, Rose flower, Peacock, elephant, etc.

- vi. Optional value:** It includes the potentials of biodiversity that presently unknown and need to be explored.

HOT SPOTS OF BIODIVERSITY

Endemic species: *The species which are confined to a particular locality only are called endemic species.*

Hotspots: *The area or region which possesses endemic species (species which are found only in a particular area) is called hot spot.*

Conditions for Hot spots:

- The site should already be threatened
- It should include a diverse range of habitats
- It should have a significant number of specialist species

Reason for rich biodiversity in tropics:

The tropics have a more stable climate.

Warm temperatures and high humidity in the tropical areas provide favorable conditions.

Among plants, rate of out-crossing appears to be higher in tropics.

Biodiversity Hotspots in India

1. Eastern Himalayas
2. Western Ghats

Eastern Himalayas:

Geographically these areas comprise Nepal, Bhutan and neighboring states of Northern India.

There are 35,000 plant species in the Himalayas, of which 30% are endemic.

E.g.: Rice, Banana, Citrus, Ginger etc. 63% of mammals is from Eastern Himalayas, 60% of the Indian birds are from North East.

Features of Himalayan Regions

1. Eastern Himalayas shows an ultra varied topography, which has species diversity and endemism.
2. In Sikkim, in semi-isolated area of 7298 km of 4250 plant species, 2550 (60%) are endemic.
3. In India's sector, there are 5800 plant species of which 2000 (36 %) are endemic.
4. In Nepal, there are 7000 plant species of which 500 (8 %) are endemic.
5. In Bhutan, there are 5000 plant species of which 750 (15 %) are endemic.

Western Ghats:

The area comprises Maharashtra, Karnataka and Tamilnadu. Nearly 1500 endemic plant species are found from western ghats, 62% of amphibians and 50% lizards are endemic. It is reported that only 6.8% of the original forests are existing today while the rest has been deforested or degraded.

Some common plants – Ternstroemia, Japonica etc.

Some common animals – Blue bird, Lizard and Hawk etc.

In India, Eastern Himalaya and Western Ghats are considered as hot spot. In world 12 hot spots contain 34,400 endemic plant species in all tropical forest.

THREATS TO BIODIVERSITY:

Any disturbance in a natural ecosystem tends to reduce its biodiversity. The waste generated due to increase of human population and industrialization spoils the environment. It leads to more diversity in biological species.

It is mainly due to,

- A) Habitat loss
- B) Poaching of wildlife
- C) Man-Wildlife Conflicts

A) HABITAT LOSS:

Definition: The loss of population of interbreeding organism is caused by habitat loss. It threatens wide range of plants and animals.

Factors influencing Habitat loss:

Deforestation: Forest and grasslands have been cleared for agricultural land and some developmental projects.

Destruction of wetlands: The wetlands are destroyed due to draining, filling and population, which causes huge diversity.

Raw Materials: For the production of hybrid seeds, the wild plants are used as raw materials.

Production of Drugs: Many pharma industries use the plants as raw materials for drug.

Developmental activities: Construction of dams near the forest, discharge of effluents kills the birds and many aquatic organisms.

Case Study 1: Habitat Loss in the Western Ghats

Location:

Western Ghats (Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu)

Issue:

The Western Ghats, one of the world's biodiversity hotspots, is experiencing severe deforestation and habitat fragmentation due to human activities.

Causes:

- Expansion of plantations (tea, coffee, rubber)
- Mining for bauxite and iron ore (especially in Goa)
- Hydroelectric and dam projects

- **Urbanization** and road building

Impact:

- Endangered species like the **Lion-tailed macaque** and **Nilgiri Tahr** lose forest cover and connectivity.
- Fragmented habitats **reduce genetic diversity**.

Conservation Efforts:

- **Establishment of protected areas** (e.g., **Silent Valley National Park**)
- Recommendations by the **Gadgil and Kasturirangan Committees**

B) POACHING OF WILDLIFE:

Definition: *Poaching means **killing of animals** or commercial hunting.*

i) Subsistence Poaching: Killing animals to **provide enough food for their survival** is called subsistence poaching.

ii) Commercial poaching: Hunting and killing of animals to **sell their products** is called commercial poaching. E.g., **Elephants killed for ivory; Star turtles: medical values; Male gorilla: Hunted for its body parts and hands.**

Factors influencing Poaching:

- Increasing in human population **increases the need of forest resources** which degrades the wildlife habitats.
- Increasing of **illegal trade, smuggling of wildlife body parts** leads to increase the poaching.

Remedial measures:

- **Illegal hunting and trade** should be stopped immediately.
- **Leather coat, purse, bag purchasing** should be avoided.
- **Biodiversity laws** to be strengthened.

Case Study 1: Tiger Poaching in Sariska Tiger Reserve

Location: Sariska Tiger Reserve, Rajasthan

Species Affected: **Bengal Tiger (*Panthera tigris tigris*)**

Incident: By **2005**, Sariska—a key Project Tiger reserve—**lost all its tigers** due to rampant poaching.

Causes:

- **Organized poaching rackets** supplying tiger parts (skin, bones) to the illegal wildlife trade.
- **Lack of monitoring and patrolling** in the forest area.
- **Involvement of local communities** in poaching due to poverty and lack of awareness.

C) MAN - WILDLIFE CONFLICTS:

Man – Wildlife conflicts arise, when wild life stats causing immense damage and danger to the man. Under such conditions it is very difficult for the forest department to compromise the affected villagers and to gain the support for wildlife conservation.

E.g.

- Sambalpur at Orissa – 195 people were killed by elephants and villages are killed 98 elephants.
- Border of kote - Chamarajanagar, Mysore several elephants were killed by human, because of massivedamage of farmer's cotton and sugar cane crops.

Factors influencing Man-Wildlife conflicts:

- Human encroachment of forest areas.
- Injured animal's attack.
- Forest department stops the practice for cultivating sugarcane paddy, coconut trees near the forest.
- Often villagers put electric wiring.

Remedial measures:

- Compensation scheme to be started for affected villagers.
- Cropping pattern should be changed near the forest areas.
- Food, water to be made available in forest zones.
- The development and constructional activities around forest area should be stopped.

Case Study 1: Human-Elephant Conflict in Assam

Location: Sonitpur and Goalpara districts, Assam

Species Involved: Asian Elephant (*Elephas maximus*)

Nature of Conflict: Frequent crop raiding, property destruction, and human deaths due to elephants straying into villages near forest fringes.

Causes:

- Encroachment on traditional elephant corridors for agriculture and tea plantations.
- Deforestation and habitat fragmentation due to logging and development.
- Elephants attracted to crops like paddy, banana, and sugarcane.

Impact:

- Dozens of human and elephant deaths annually.
- Retaliatory poisoning or electrocution of elephants by villagers.
- Rise in stress and aggression among elephant herds.

Mitigation Efforts:

- Use of solar-powered fences and chili smoke bombs to deter elephants.
- Creation of elephant corridors to restore migration routes.
- Community awareness and involvement through eco-development committees (EDCs).

INDIA AS A MEGA-DIVERSITY NATION

- India is a mega-diverse nation because it contains a disproportionately large number of Earth's species within a small percentage of its land area.
- The country harbors 7-8% of all recorded species while occupying only 2.4% of the world's land, with significant numbers of both flora and fauna, including many endemic species.
- Key contributing factors to India's rich biodiversity include its varied climates, diverse ecosystems, and the presence of four global biodiversity hotspots: the Himalayas, Indo-Burma, Western Ghats, and Sundaland.

Examples of Biodiversity

Plants: Over 45,500 species of plants have been documented, with traditional knowledge of many native plants used in indigenous healing practices.

Animals: India has rich faunal diversity, with high numbers of endemic vertebrate groups, including many species of amphibians and reptiles.

Threatened Species: The Malabar giant squirrel is an example of an endemic species found in the Western Ghats, facing threats from habitat loss.

Conservation Efforts

Protected Areas: India has established numerous protected areas, such as national parks, wildlife sanctuaries, and 18 biosphere reserves, to conserve its natural heritage.

International Conventions: The country's biodiversity conservation efforts are supported through participation in international conventions, highlighting the importance of protecting ecosystems and genetic resources.

ENDANGERED SPECIES:

The species is said to be endangered, when **its number has been reduced to a critical level**. Unless it is protected and conserved, it is in **immediate danger of extinction**.

Endangered species In India,

450 plants

150 birds

100 mammals

25 reptiles

3 fishes

50 insects

E.g. Tiger, Black Rhinoceros, Giant Panda, Himalayan black bear, etc. Many medicinal plants are also found as the endangered species in India such as Serpentina which has many medicinal values.

Mammals

Species	Scientific Name	Habitat
Bengal Tiger	<i>Panthera tigris tigris</i>	Forests (Pan India)
Asiatic Lion	<i>Panthera leo persica</i>	Gir Forest, Gujarat

Birds

Species	Scientific Name	Habitat
Great Indian Bustard	<i>Ardeotis nigriceps</i>	Rajasthan, Gujarat grasslands

RED DATA BOOK: *It contains the list of endangered species of plants and animals. It gives the warning signal for those species which are endangered and if not protected they become extinct in near future.*

The **Red Data Book** is a document that provides information on the **conservation status of plant and animal species** that are **facing the risk of extinction**.

Published by:

- **International Union for Conservation of Nature (IUCN)**

Purpose:

- To record and highlight **species that are endangered, vulnerable, extinct, or critically endangered**.
- To help in **conservation planning and awareness**.

Importance:

- Acts as a **global reference** for threatened species.
- Helps governments and conservationists prioritize efforts.
- Encourages **international cooperation** for species protection.

Indian Version:

India also maintains a **Red Data Book** through the **Zoological Survey of India (ZSI)** and **Botanical Survey of India (BSI)** for native species.

Factors Affecting Endangered Species

Endangered species are those at risk of extinction due to various **natural and human induced factors**.

1. Habitat Loss and Degradation

- Destruction of forests, wetlands, and grasslands for **agriculture, urbanization, and industries**.
- Leads to loss of shelter, food, and breeding areas.

2. Poaching and Illegal Wildlife Trade

- Animals hunted for **skin, bones, horns, tusks**, and other body parts.
- Common in tigers, rhinos, elephants, etc.

3. Pollution

- **Water, air, and soil pollution** affect the health and reproduction of species.
- Chemical pesticides harm birds, aquatic life, and amphibians.

4. Climate Change

- Rising temperatures, erratic rainfall, and melting glaciers disturb ecosystems.
- Species like polar bears and coral reefs are severely affected.

5. Invasive Species

Non-native species introduced by humans often **compete for resources**, spread diseases, or prey on native species.

6. Overexploitation

Overfishing, overgrazing, and excessive logging reduce population numbers beyond recovery.

7. Fragmentation of Habitat

Infrastructure like **roads, dams, and railways** divide natural habitats, affecting animal movement and breeding.

These factors must be addressed through **conservation efforts, laws, and awareness** to protect endangered species.

ENDEMIC SPECIES

The species which are found only in a particular region are known as endemic species. 47,000 species, 7000 plants are endemic in India. Nearly 62% of the species are available in Himalayas and Western Ghats.

Fauna : Animals present in a particular region or period. The Western Ghats are particularly rich in amphibians and reptiles. 62 % of amphibians and 50 % of reptiles are endemic in Western Ghats.

Flora: Plants present in particular region or period. Nearly 200 Pteridophyta and 4750 angiosperms are endemic. In general, in Land 878 species, Fresh water 89 species are endemic.

Endemic Mammals of India

Common Name	Scientific Name	Region
Nilgiri Tahr	<i>Nilgiritragus hylocrius</i>	Western Ghats
Lion-tailed Macaque	<i>Macaca silenus</i>	Western Ghats
Malabar Civet	<i>Viverra civettina</i>	Kerala (Critically Endangered)
Indian Giant Squirrel (Malabar Squirrel)	<i>Ratufa indica</i>	Central & South India
Kashmir Stag (Hangul)	<i>Cervus hanglu hanglu</i>	Kashmir Valley
Andaman Horseshoe Bat	<i>Rhinolophus cognatus</i>	Andaman Islands

Factors affecting endemic species:

1. Habitat Loss

- **Deforestation**, urbanization, and agriculture destroy natural habitats.
- Especially dangerous for species restricted to **forests, wetlands, or islands**.

2. Climate Change

- Changes in temperature and rainfall affect **sensitive ecosystems**.
- Alters the **microclimates** that endemic species depend on.

3. Invasive Species

- Introduction of non-native plants or animals can **outcompete, prey upon**, or bring **diseases** to endemic species.

4. Pollution

- Air, water, and soil pollution degrade habitats and **harm species health**.
- Especially harmful to **amphibians, aquatic life, and plants**.

5. Over exploitation

- Overgrazing, hunting, or collecting endemic plants and animals for **trade or consumption** reduces their numbers drastically.

6. Natural Disasters

- Earthquakes, landslides, cyclones, or volcanic eruptions can **wipe out entire populations**, especially on islands or small habitats.

7. Limited Range and Low Population

- Being restricted to a small area makes endemic species **less adaptable** and more prone to extinction if conditions change.

CONSERVATION OF BIODIVERSITY

Conservation of biodiversity is the practice of protecting, preserving, and managing the variety of life on Earth including plants, animals, and microorganisms and their ecosystems to ensure their long-term survival and the sustainable use of natural resources for current and future generations

I.) In-Situ Conservation of Biodiversity:

In-Situ conservation involves conservation of species within its natural habitat – places where the species normally occurs. In this method biodiversity conserved where it currently exists.

Methods:

1) **Biosphere Reserve:** In India, 18 biosphere reserves are available. The nature or biosphere reserve covers large area more than 5000 sq. km, where human activities and tourism is permitted.

The aim of biosphere reserve:

- To give long term survival of evolving ecosystem
- To conserve maximum number of species and communities
- To protect endangered species
- To serve as sites of recreation and tourism
- Nandadevi - U.P, Manas - Assam, Nilgiri - TN etc.

2) **National parks:** In India 120 National parks are available. They are small reserves covering an area about 100 sq. km to 500 sq. km and protecting the species and their habitats. The boundaries of national parks are well defined. But tourism is permitted under certain conditions.

Name of National Park	State	Wildlife
Kaziranga	Assam	One horned Rhino
Gir National Park	Gujarat	Indian Lion
Periyar	Kerala	Tiger and Elephant
Mudumalai	Tamil Nadu	Elephant, Tiger

3) Wild life Sanctuaries: A wild life sanctuary is an area, which is reserved for the conservation of animals only. At present, there are 515 wildlife sanctuaries in our country. It provides good ecosystem for animals. Killing, hunting, shooting and capturing of animals are prohibited.

Name of Sanctuary	State	Major Wildlife
Ghana Bird Sanctuary	Bihar	Tiger and Leopard
Sultanpur Bird Sanctuary	Haryana	Migratory Bird
Vedanthangal	Tamil Nadu	Water Birds

Gene Sanctuary:

A gene sanctuary is an area for the conservation of plants only.

- One gene sanctuary for citrus (Lemon Family)
- Gene sanctuary for pitcher plant (an insect eating plant)

Other projects:

- Project Tiger
- Gir Lion Project
- Project Elephant

Advantages:

- 1) It is very cheap and convenient method
- 2) Species adjusted to the natural disasters.

Disadvantages:

- 1) Large surface area is required
- 2) Improper maintenance due to shortage of staff.

II.) Ex-Situ conservation of Biodiversity:

It involves the protection of fauna and flora outside the natural habitat. The Maintenance of endangered plant and animal species under controlled conditions.

Methods:

1) National Bureau of Plant Genetic Resources (NBPGR):

It is located in New Delhi, it uses **cryo preservation** technique to preserve agricultural and horticultural crops. It involves the preservation of seeds, crop by using liquid nitrogen at a temperature less than -196° C. Variety of rice, radish, tomato, onion, chilli etc. is preserved for several years.

2) National Bureau of Animal Genetic Resources (NBAGR):

It is located in Karnal, Haryana. It preserves the semen of domesticated bovine animals. (Cow related)

3) National Facility for Plant Tissue Culture Repository (NFPTCR): It develops the facility for conservation of varieties of crop plants or trees by tissue culture. This facility has been created within the NBPGR.

Advantages:

- It the best of method of conservation of endangered species.
- It conserves the population of endangered species.

Disadvantages:

- It is very expensive.
- As the wildlife population is more, it is very difficult to conserve.

Part – A

1. Define the term “Biodiversity”.

The variety and variability among all group of organisms and the ecosystem in which they occur is called biodiversity.

2. State the significance and scope of environmental education.

- To get an awareness and sensitivity to the total environment and its allied problems.
- Acquiring basic understanding of total environment problems and the responsible presence of humanity in it.
- To motivate active participation in environmental protection and improvement.

3. Define environmental science.

The study of basic components of our surroundings and their interaction is called environmental science.

4. Enumerate the human activities which destroy the biodiversity.

a) Unplanned urbanization b) Industrialization c) Dam construction d) Deforestation

5. Define the terms producers and consumers.

Producers (Autotrophs): Those organisms producing their own food through photosynthesis are called producers. Ex. Plants, algae, etc

Consumers (Heterotrophs): Those organisms feeding on other organisms are called consumers.

Ex. Deer, Cow, Jackal, Wolf, Lion, etc.

6. What is meant by genetic diversity?

Within a species there are number of varieties which are slightly different from one another due to their genetic code. This is called genetic diversity.

7. Distinguish between primary and secondary consumer.

Primary consumers: Plant eating animals are called herbivores. Ex. Deer, Cow, etc.

Secondary consumers: Meat eating animals are called carnivores. Ex. Jackal, Wolf, Lion, etc.

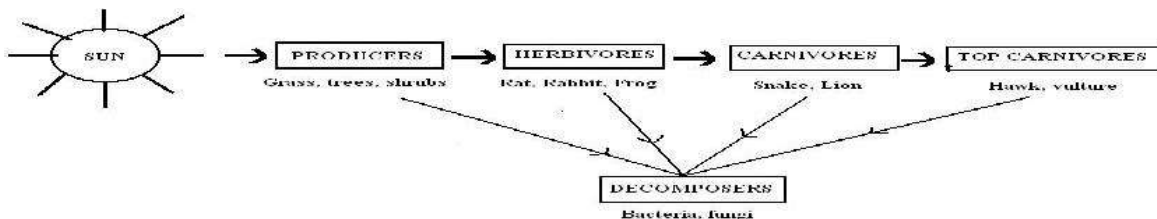
8. Explain ecological succession.

The sequential changes of community over a period of time in the same area are called ecological succession.

9. Write any four functions of forests.

It brings rainfall b) It prevents flood c) It provides food and shelter for animals d) It provides valuable products.

10. Enumerate the structural and functional elements of ecosystem (OR) Give flow chart of energy flow in an ecosystem.



11. What are the classifications of biotic components of ecosystems?

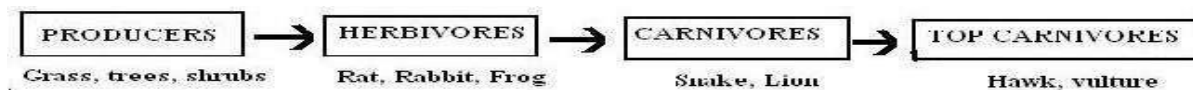
Producers (Autotrophs): Those organisms producing their own food through photosynthesis are called producers. Ex. Plants, algae, etc

Consumers (Heterotrophs): Those organisms feeding on other organisms are called consumers. Ex. Deer, Cow, Jackel, Wolf, Lion, Human being, crow, etc.

Decomposers (Reducers): Those organisms break down complex compounds of dead organisms are called decomposers. Ex. Bacteria and fungi.

12. What is food chain and food web?

The sequence of eaters and being eaten is called food chain.



“In an ecosystem, various food chains are inter connected with each other to form a net work is called food web”



13. What is ecosystem?

The system in which group of biotic and abiotic components are inter-related one another is called ecosystem. Ex. Forest ecosystem, marine ecosystem, etc.

14. Explain threatened and endangered species.

- Those species which are vulnerable to endangerment in the near future are called threatened species. Ex. African grey parrot, American paddlefish, hippopotamus etc.
- The species is said to be endangered, when its number has been reduced to a

critical level. Unless it is protected, it is in immediate danger of extinction.

Example: Tiger, Black Rhinoceros, Giant Panda etc.

15. Define ecological succession and mention its types.

The progressive replacement of one community by another by another till the development of a stable community in a particular area is called ecological succession.

The types are primary succession and secondary succession.

16. What are the two important bio-diversity hot spots in India?

i. Eastern Himalayas 2) The Western Ghats

17. Name any two extinct species of India.

Tiger, Black Rhinoceros, Giant Panda, Himalayan black bear, Egyptian land tortoise, etc.

18. Write the role of decomposers in ecosystem.

Those organisms break down complex compounds of dead organisms are called decomposers.

Ex. Bacteria and fungi.

19. Give the causes for endangerment of species.

- A) Habitat loss due to deforestation
- B) Poaching of wildlife for their valuable properties
- C) Man-Wildlife Conflicts like urbanization and industrialization

20. What is red data book?

RED data book contains the list of endangered species of plants and animals. It gives the warning signal for those species which are endangered and if not protected they become extinct in future.

QUESTION BANK

Part – B

- 1) Explain the structure, function and energy flow of an ecosystem.
- 2) Explain in detail about Ecological succession
- 3) Explain the types and values of biodiversity.
- 4) Discuss the hotspots of biodiversity and threat of biodiversity.
- 5) Discuss in detail: Conservation of biodiversity.
- 6) Explain India is a mega diversity nation.
- 7) Write a note on endangered and endemic species
- 8) What do you understand by the hot spots of biodiversity that extend in India.